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Computer Science and Engineering Department
M.Tech(CSE/DS)-I - Sem-I - 2026-27
Computer Vision and Image Processing (CSCS111/CSDS119)
Lab Assignment 5: Spatial and Frequency Domain Filtering

PART A : Spatial and Frequency Domain Filtering

Note: For this experiment, the Cameraman image will be used.

1. Spatial Domain Filtering

Write a program to demonstrate the working of spatial domain filtering techniques for image smoothing and sharpening. Apply mean, Gaussian, and Laplacian filters to the Cameraman image. Experiment with different neighborhood sizes and compare the results with respect to noise reduction, smoothing, edge enhancement, and preservation of image details. Write your observations.

2. Discrete Fourier Transform (DFT)

The DFT of an image $f(x, y)$ is defined as:

$$F(u, v) = \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f(x, y) e^{-j2\pi \left(\frac{ux}{M} + \frac{vy}{N} \right)}$$

Compute the Discrete Fourier Transform (DFT) of a grayscale image and visualize its frequency domain representation. Compute the 2D DFT, display the magnitude spectrum, and perform the Inverse DFT to reconstruct the original image. Also, visualize and compare the magnitude and phase spectra of the image.

3. Ideal Low-Pass Filter (ILPF)

Define and apply an ideal low-pass filter in the DFT domain. Define a circular low-pass filter, apply it to the DFT of the image, and perform the inverse DFT to obtain the filtered image. Display the magnitude spectrum of the filter, the original image, and the filtered image. Try different combinations of D_0 and write your observations regarding smoothing and loss of image details.

4. Ideal High-Pass Filter (IHPF)

Define and apply an ideal high-pass filter in the DFT domain. Apply the filter to the DFT of the image and perform the inverse DFT to obtain the filtered image. Try different combinations of D_0 and display the magnitude spectrum of the filter, the original image, and the filtered image. Write your observations regarding edge and detail enhancement.

5. Gaussian Low-Pass and High-Pass Filters

Define and apply a Gaussian low-pass filter (GLPF) and Gaussian high-pass filter (GHPF) in the DFT domain. Perform the inverse DFT to obtain the filtered images and experiment with different values of D_0 . Compare the results with the ideal filters and write your observations.

6. Butterworth Low-Pass and High-Pass Filters

Define and apply Butterworth low-pass (BLPF) and Butterworth high-pass filters in the DFT domain. Experiment with different values of D_0 and filter order n . Compare the results obtained using ideal, Gaussian, and Butterworth filters and write your observations.

Note: Assignment can be done in a group of maximum size of two. For reading and writing images you may use built-in functions from OpenCV or any other library you use.

PART B : Exploratory Problem

1. Cross-Correlation for Template Matching

Suppose that you are a manager at a grocery store. One of your responsibilities is to check the shelves periodically and stock them up whenever there are sold-out items. You got tired of this laborious task and decided to build a computer vision system that keeps track of the items on the shelf.

Luckily, you have learned in CVIP that cross-correlation can be used for template matching: a template g is multiplied with regions of a larger image f to measure how similar each region is to the template. The template of a product (`template.jpg`) and the image of shelf (`shelf.jpg`) is provided. Use cross-correlation to find the product in the shelf and mark the detected location in the result using a blue rectangle. Write your observation about the detected location and the matching result.